RESEARCH PAPER

Emergency Savings and Household Financial Fragility

Evidence from SHED Published Indicators, 2013–2025

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DOCUMENT TYPE
Research paper

DATE
May 2026

Abstract

This paper studies household emergency-payment capacity using published public indicators from the Federal Reserve Survey of Household Economics and Decisionmaking (SHED). It focuses on two recurring adult-level measures: whether respondents would cover a hypothetical $400 emergency expense completely using cash or its equivalent, and whether respondents report emergency savings sufficient to cover three months of expenses. The paper is intentionally descriptive. It summarizes public table values and provides a measurement frame; it does not claim causal identification, estimate household-level regressions, or observe bank balances.

The central published-indicator fact is a plateau after a temporary 2021 improvement. In 2025, 63 percent of adults said they could cover a $400 emergency expense with cash or its equivalent, leaving 37 percent who could not. Fifty-five percent reported three months of emergency savings, leaving 45 percent without that larger reserve. Both measures improved through 2021 and then partially reversed. The $400 cash/equivalent payment-capacity measure rose from 63 percent in 2019 to 68 percent in 2021, then returned to 63 percent by 2025. The three-month savings measure rose from 53 percent in 2019 to 59 percent in 2021, then fell to 55 percent by 2025.

The distribution is more informative than the average. In 2025, 74 percent of adults with less than a high school degree lacked the $400 cash/equivalent payment capacity, compared with 19 percent of adults with a bachelor's degree or more. Sixty percent of Black adults and 54 percent of Hispanic adults lacked that payment capacity, compared with 27 percent of White adults and 23 percent of Asian adults. Fifty-five percent of adults ages 18-29 lacked the $400 cash/equivalent payment capacity, compared with 22 percent of adults ages 60 and older. These unconditional gaps should not be read causally. They show where published SHED indicators locate emergency-payment fragility and motivate a microdata research design.

1. Introduction

A household's resilience depends not only on whether income is positive, debt is current, or employment is available. It also depends on whether the household has liquid resources at the moment a shock arrives. A car repair, medical bill, utility arrear, missed shift, insurance deductible, rent increase, or family emergency can become a crisis if the household has no liquid buffer.

This paper studies that buffer directly. The paper was written as part of a broader sequence on household exposure to monetary and cost pressure, but this draft should stand on a narrower claim: published SHED indicators show that emergency-payment capacity improved through 2021, then returned to roughly its pre-2021 level, while large unconditional gaps remain across socioeconomic groups. The question is simple: when a disruption arrives, who appears to have a shock absorber?

The evidence comes from the Federal Reserve's SHED public tables. The paper focuses on two measures. The first is immediate liquidity: whether an adult would cover a hypothetical $400 emergency expense completely using cash or its equivalent. The second is a larger precautionary reserve: whether an adult has emergency savings to cover three months of expenses. These measures are not perfect. The $400 threshold is nominal and fixed over time, so its real value declines as prices rise. The three-month measure is self-reported and does not verify account balances. Both are adult-level survey measures, not household balance sheets. But they are recurring, transparent, distributional, and directly tied to financial fragility.

The paper's claim is descriptive. It does not estimate a causal effect of inflation, monetary policy, credit conditions, pandemic transfers, or fiscal withdrawal on emergency savings. It shows that published emergency-buffer indicators improved into 2021, then deteriorated or plateaued afterward, and that unconditional gaps remain large across education, race and ethnicity, age, parental status, and metro status.

2. Conceptual framework: liquidity as the household shock absorber

The emergency buffer is the liquid layer between a household and distress. A simple accounting expression is:

$$B_{ht} = L_{ht} / E_{ht}$$

where B_{ht} is the household buffer ratio, L_{ht} is liquid resources available quickly without high-cost borrowing or forced asset sales, and E_{ht} is the relevant emergency expense or near-term expense requirement. A household with $B_{ht} >= 1$ can absorb the shock using liquid resources. A household with $B_{ht} < 1$ must borrow, sell assets, delay payment, seek help, or fail to pay.

The $400 SHED measure is a small-shock version of this concept. More precisely, it asks whether an adult would cover a common expense using cash or its equivalent; SHED wording treats cash, savings, or a credit card paid off at the next statement as cash-equivalent payment capacity. The three-month savings measure is a larger-shock version. It asks whether the household has a reserve for sickness, job loss, economic downturn, or other emergencies.

This framework links directly to financial-fragility research. A household can have income but lack liquidity. It can own illiquid assets but still be unable to cover a near-term expense. It can be current on bills but one shock away from expensive credit. The published emergency-buffer indicators therefore point to something different from net worth, income, or employment, while remaining imperfect proxies for true liquid balance-sheet capacity.

It also clarifies the connection to resilience costs. If the cost of staying less fragile rises, emergency-payment capacity determines whether households can pay that cost without becoming more fragile. The two margins interact: recurring resilience costs absorb cash flow, and weak liquidity makes those costs more destabilizing.

3. Data and measurement

The paper uses public Federal Reserve SHED data visualization table values, last updated May 13, 2026. The unexpected-expense table reports the share of adults who would cover a $400 emergency expense completely using cash or its equivalent. The emergency-savings table reports the share of adults who have emergency savings to cover three months of expenses.

The $400 measure is available for all adults from 2013 through 2025, with breakdowns by education, race and ethnicity, age, metro/non-metro status, and parental status. The three-month emergency-savings measure is available from 2015 through 2025 with similar breakdowns. The analysis transcribes the published public table values into machine-readable local CSV files and then generates tables and figures. The Federal Reserve also publishes SHED public-use CSV microdata for each survey year, including 2025, and those files are the appropriate source for the next-stage inferential version of the project.

Several limitations are important. First, the $400 threshold is not inflation-adjusted. Because $400 in 2013 is not the same real burden as $400 in 2025, a flat series over time may understate deterioration in real emergency capacity. Second, the measure asks how respondents would pay a hypothetical expense; it does not directly observe bank balances. Third, “cash or equivalent” includes cash-like payment capacity, including a credit card paid off at the next statement in SHED wording. Fourth, the three-month savings measure is self-reported and depends on respondents' own expense baseline. Fifth, the public tables are aggregates by characteristic; this paper does not estimate multivariate microdata models, design-corrected standard errors, confidence intervals, decompositions, or outcome-prediction models. Sixth, the public table values are rounded, so small one- or two-point movements should not be overinterpreted without survey-adjusted inference.

Those limitations do not make the measures uninformative. They mean the results should be read as transparent public indicators of adult emergency-payment capacity, not as complete household balance sheets or causal estimates.

4. Time pattern: improvement, reversal, plateau

Table 1 shows the central time pattern. The $400 cash/equivalent payment-capacity measure improved from 50 percent in 2013 to 63 percent in 2019, then to 68 percent in 2021. By 2022 it fell back to 63 percent and remained at 63 percent through 2025. The complement is the fragility measure: in 2025, 37 percent of adults could not cover a $400 expense with cash or its equivalent.

The larger three-month measure follows a similar pattern. It rose from 47 percent in 2015 to 53 percent in 2019 and 59 percent in 2021. It then declined to 54 percent in 2022 and stood at 55 percent in 2025. In other words, 45 percent of adults lacked three months of emergency savings in 2025.

Table 1. Emergency-buffer milestones, all adults. Values are percent of adults. The $400 measure is ability to cover the expense completely using cash or its equivalent. The 3-month measure is having emergency savings to cover three months of expenses.

Year	$400 cash/equivalent	Cannot cover $400 with cash/equiv.	Has 3-month emergency savings	Lacks 3-month emergency savings
2013	50	50	—	—
2015	54	46	47	53
2019	63	37	53	47
2021	68	32	59	41
2022	63	37	54	46
2024	63	37	55	45
2025	63	37	55	45

Figure 1 shows the $400 cash/equivalent payment-capacity trend. The all-adults series improved through 2021 and then returned to the pre-2021 level. The education gradient is persistent. Adults with a bachelor's degree or more remain far more likely to have the small emergency buffer than adults with less than a high school degree.

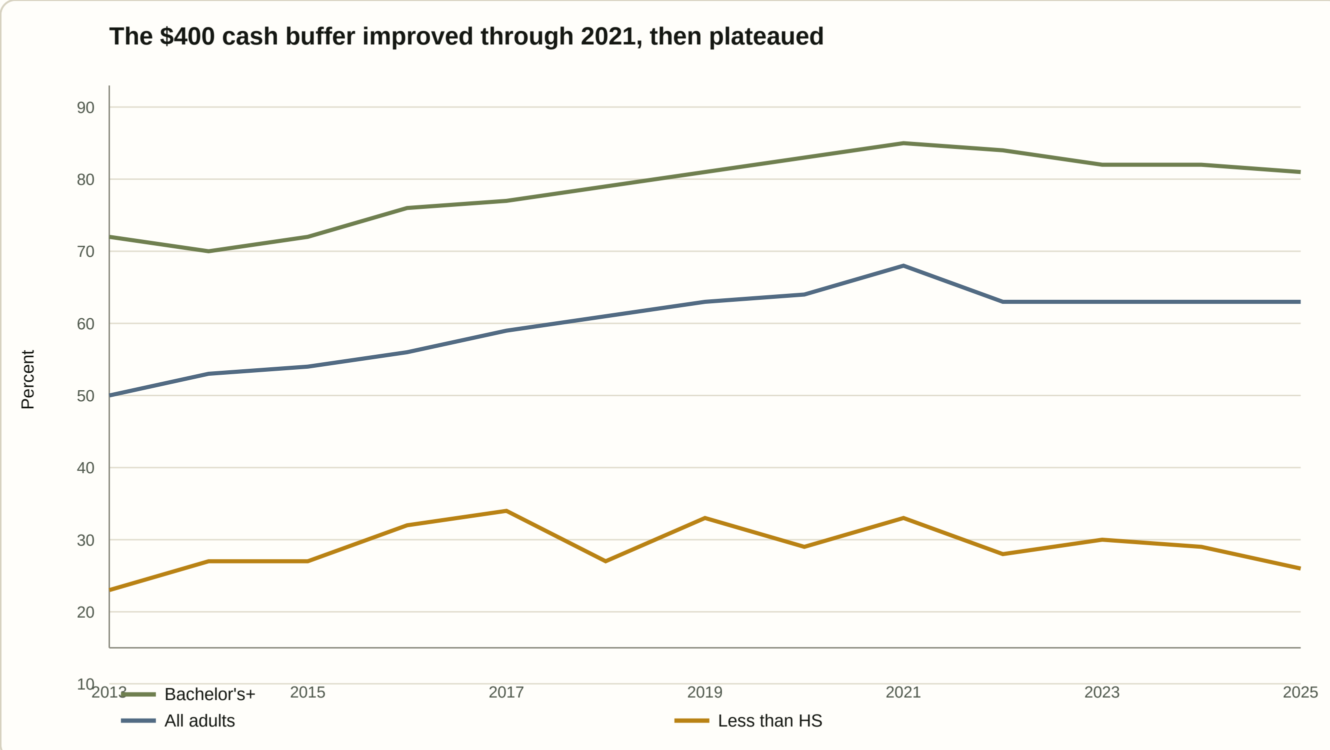

Figure 1. The $400 cash/equivalent payment-capacity indicator improved through 2021, then plateaued.

Figure 2 shows the three-month emergency-savings trend. This larger buffer also peaked in 2021 and then weakened. Parents living with their own children under age 18 consistently report lower emergency savings than all other adults, which is consistent with the idea that caregiving and family costs absorb liquid slack.

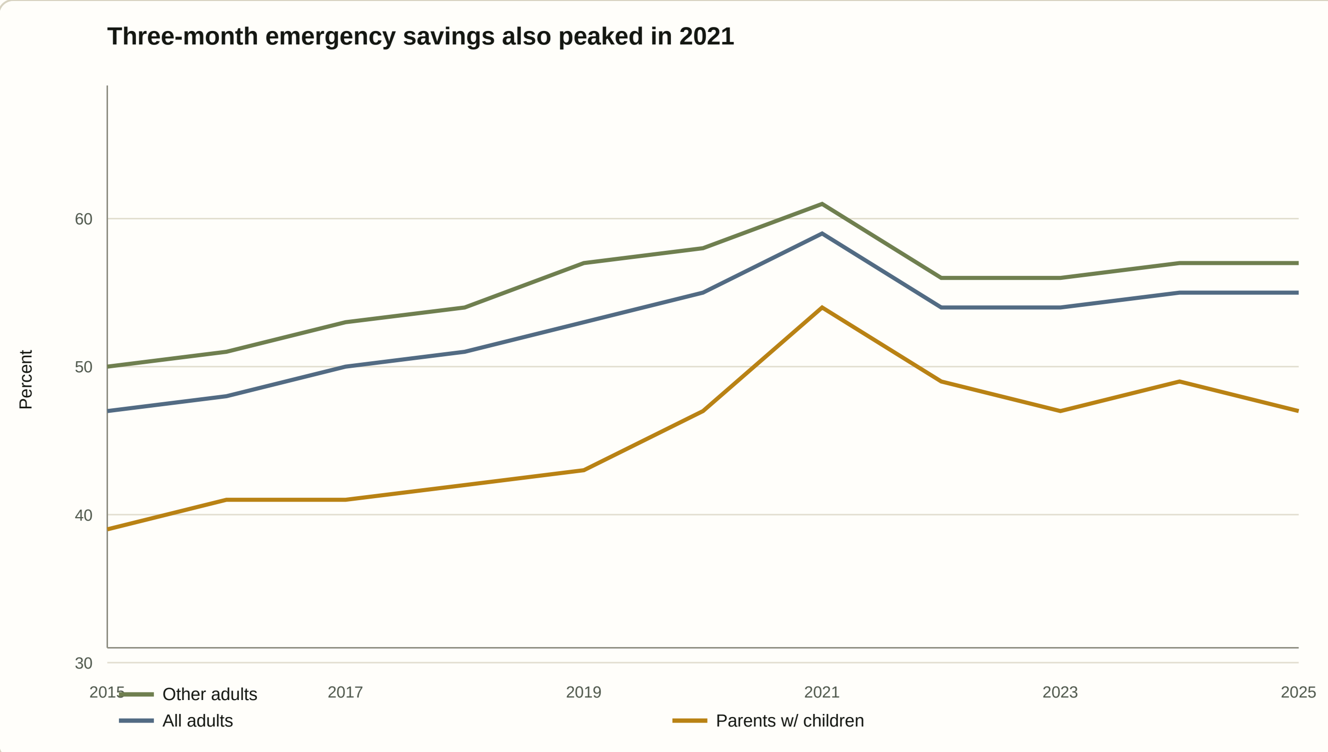

Figure 2. Three-month emergency savings also peaked in 2021.

5. Distribution in 2025

Table 2 shows why the emergency-buffer question is inherently distributional. The all-adults average hides sharp differences by education, race and ethnicity, age, parental status, and metro status.

Education gaps are especially large. In 2025, 74 percent of adults with less than a high school degree lacked the $400 cash/equivalent payment capacity, compared with 19 percent of adults with a bachelor's degree or more. For three-month emergency savings, 79 percent of adults with less than a high school degree lacked the buffer, compared with 28 percent of adults with a bachelor's degree or more.

Race and ethnicity gaps are also large. In 2025, 60 percent of Black adults and 54 percent of Hispanic adults lacked the $400 cash/equivalent payment capacity, compared with 27 percent of White adults and 23 percent of Asian adults. For three-month emergency savings, 62 percent of Black adults and 57 percent of Hispanic adults lacked the buffer, compared with 39 percent of White adults and 32 percent of Asian adults.

Age gaps point in the same direction. Fifty-five percent of adults ages 18-29 lacked the $400 cash/equivalent payment capacity in 2025, compared with 22 percent of adults ages 60 and older. Sixty-three percent of adults ages 18-29 lacked three-month emergency savings, compared with 29 percent of adults ages 60 and older.

Table 2. Emergency-buffer distribution, 2025. Values are percent of adults by characteristic. The gap columns are complements and represent adults without the stated buffer.

Characteristic	$400 cash/equiv.	Without $400 cash/equiv.	3-month savings	Without 3-month savings
Less than high school	26	74	21	79
High school or GED	51	49	42	58
Some college / technical / associate	59	41	51	49
Bachelor's degree or more	81	19	72	28
Black	40	60	38	62
Hispanic	46	54	43	57
White	73	27	61	39
Asian	77	23	68	32
18-29	45	55	37	63
30-44	57	43	49	51
45-59	66	34	55	45
60+	78	22	71	29
Parents with own children under 18	55	45	47	53
All other adults	66	34	57	43
Non-metro	59	41	49	51
Metro	64	36	56	44

Figure 3 displays the 2025 distribution for the small-shock published measure. The chart makes the central point visually: the average 37 percent without the $400 cash/equivalent payment capacity is not evenly distributed. Some groups face majority exposure.

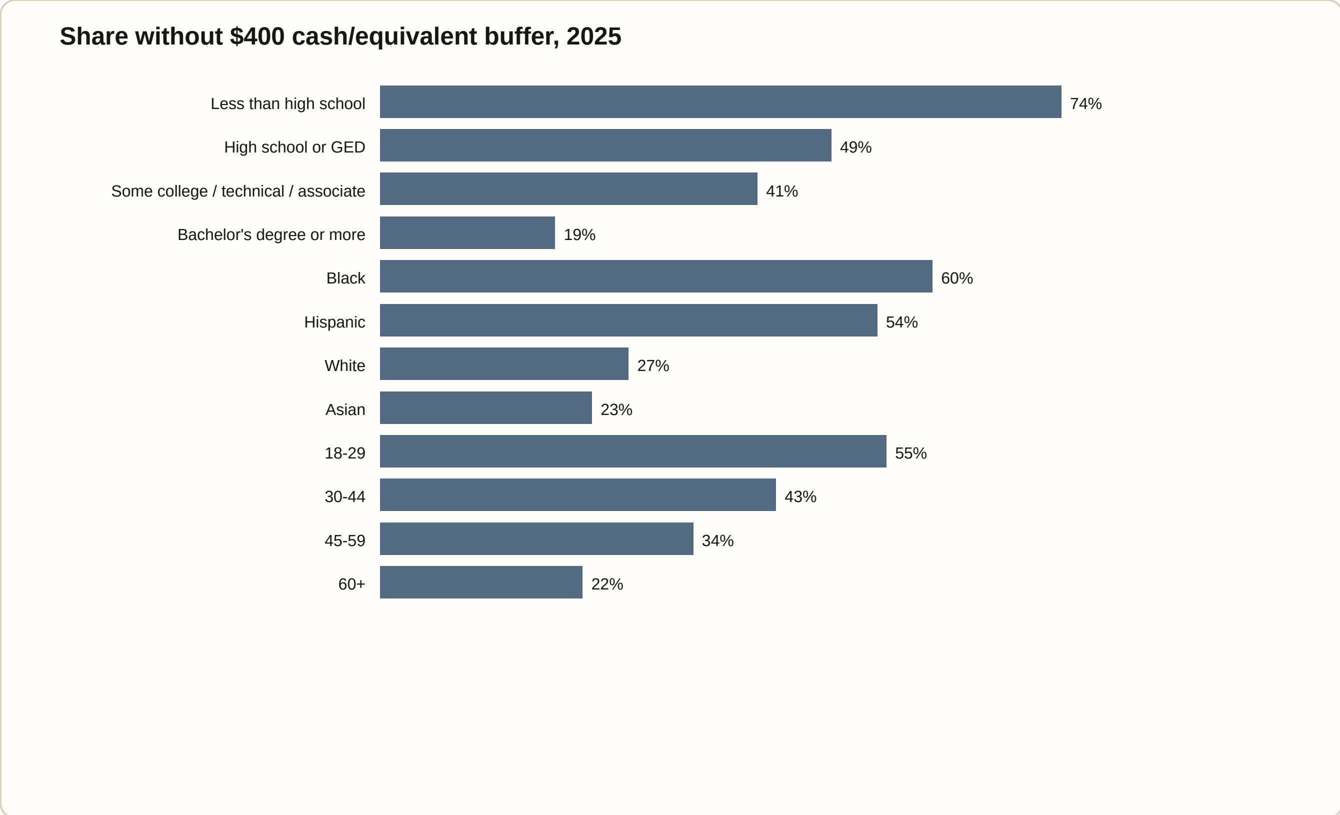

Figure 3. Share without $400 cash/equivalent buffer, 2025.

Figure 4 displays the larger three-month savings gap. The larger-shock measure is weaker for nearly every group. This is unsurprising: covering a $400 shock is easier than covering three months of expenses. But the larger measure is closer to the financial-resilience question that matters for job loss, illness, caregiving disruption, or large deductible exposure.

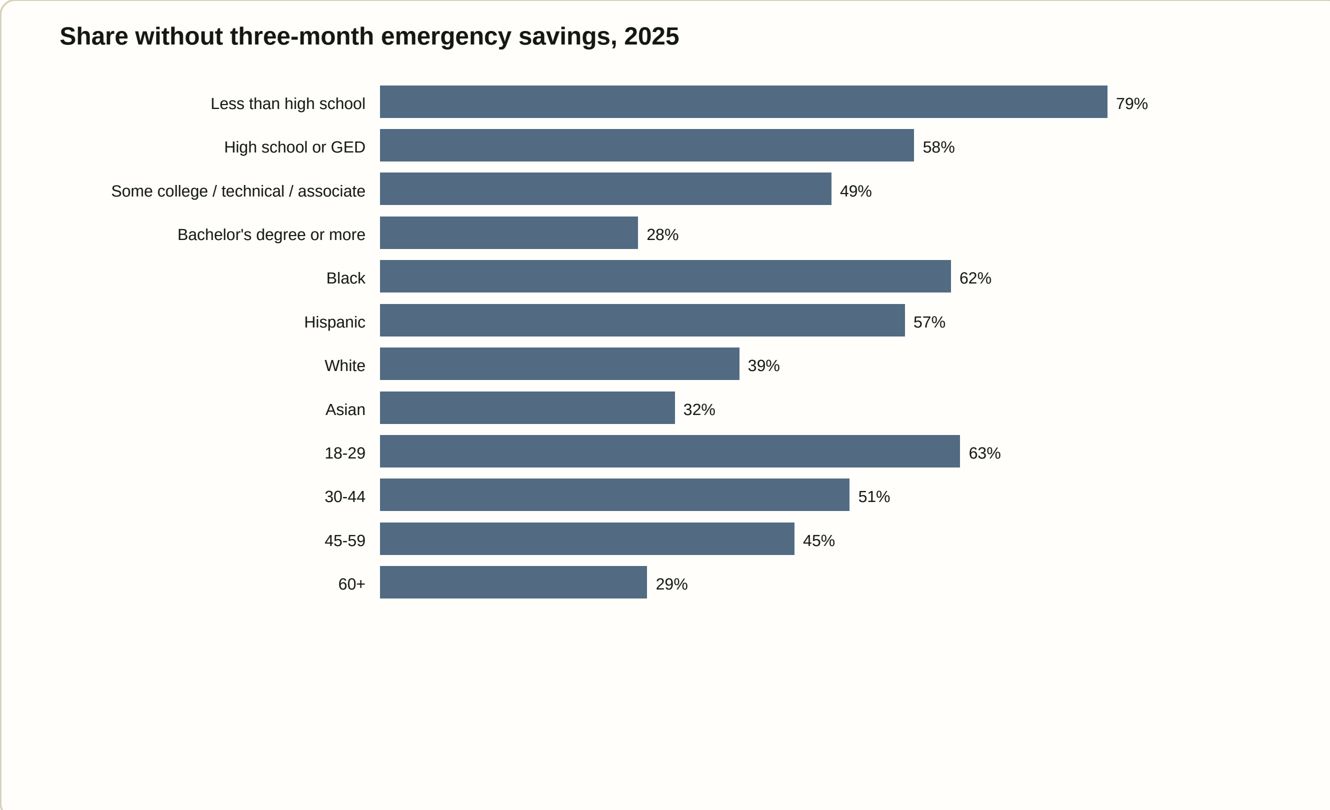

Figure 4. Share without three-month emergency savings, 2025.

6. The 2021 high and the post-2021 reversal

Table 3 measures changes around the 2021 high. For all adults, the $400 cash/equivalent payment-capacity measure improved 5 percentage points from 2019 to 2021, then fell 5 points from 2021 to 2025. The three-month savings measure improved 6 points from 2019 to 2021, then fell 4 points from 2021 to 2025.

The reversal was sharper for some groups. Adults ages 18-29 gained 8 points on the $400 measure from 2019 to 2021 and then lost 13 points by 2025. Parents living with children under age 18 gained 10 points and then lost 9. Black adults gained 5 points and then lost 8. Hispanic adults gained 9 points and then lost 8. Adults with less than a high school degree saw no improvement in the $400 measure from 2019 to 2021 and then lost 7 points by 2025.

Table 3. Emergency-buffer changes around the 2021 high. Entries are percentage-point changes. Positive values mean the buffer measure improved; negative values mean it deteriorated.

Characteristic	$400: 2019→2021	$400: 2021→2025	3-month: 2019→2021	3-month: 2021→2025
All adults	5	-5	6	-4
Less than high school	0	-7	4	-9
High school or GED	5	-5	7	-6
Some college / technical / associate	5	-6	5	-4
Bachelor's degree or more	4	-4	6	-4
Black	5	-8	6	-6
Hispanic	9	-8	12	-8
White	4	-2	5	-3
18-29	8	-13	6	-9
60+	1	-1	2	-2
Parents with own children under 18	10	-9	11	-7
All other adults	2	-3	4	-4

The interpretation should be careful. The paper does not identify why 2021 was the high point. The timing is consistent with several mechanisms--pandemic-era transfers, forced saving during restricted activity, labor-market recovery, inflation, expiring supports, and cost increases--but the public table evidence alone does not isolate any of them. What it can show is the pattern: published emergency-buffer indicators improved into 2021 and then retreated, with many vulnerable groups giving back much or all of the gain.

7. Why a fixed $400 threshold understates the issue

The $400 measure is useful because it is simple and repeated. But its fixed nominal threshold is also a weakness. A $400 expense in 2025 represents a smaller real shock than a $400 expense in 2013. Therefore, a flat ability-to-pay share over time does not necessarily mean household liquidity is equally strong. If prices rise while the threshold stays fixed, households are being asked whether they can cover a smaller real emergency.

This matters for interpreting the post-2021 plateau. The all-adults $400 cash/equivalent payment-capacity measure was 63 percent in 2019 and 63 percent in 2025. On its face, that looks like no change. But the real value of the test declined over that period. Using annual CPI-U averages from FRED, $400 in 2013 had roughly the same purchasing power as about $553 in 2025; $400 in 2019 had roughly the same purchasing power as about $504 in 2025. If the threshold had been inflation-adjusted and if the microdata permitted reconstruction of responses at higher dollar thresholds, the apparent plateau might look weaker.

The three-month savings measure partly addresses this concern because it scales to the respondent's own expenses. It asks about the ability to cover three months of expenses rather than a fixed dollar amount. That measure also weakened from its 2021 high and remained below that high in 2025. The two measures together therefore tell a stronger story than either alone: small-shock cash/equivalent payment capacity plateaued, and larger emergency savings remained incomplete for nearly half of adults.

8. Relation to existing research

This paper fits within the household financial-fragility and precautionary-saving literatures. Lusardi, Schneider, and Tufano's work on the ability to raise emergency funds emphasizes that many households are vulnerable to moderate shocks. Kaplan, Violante, and Weidner's “wealthy hand-to-mouth” framework shows why low liquid wealth can matter even when households hold illiquid assets. Related work on consumption smoothing, liquidity constraints, income volatility, household credit, and balance-sheet heterogeneity studies the mechanisms that determine whether shocks are absorbed smoothly or translated into arrears, high-cost borrowing, foregone care, and persistent distress.

The SHED measures used here are simpler than those frameworks, but they point to the same margin: liquidity is distinct from income and net worth. The present draft does not claim to extend that literature with a new identified estimate. Its contribution is narrower: it organizes recurring public SHED indicators into a transparent measurement note, quantifies the post-2021 reversal/plateau, highlights the distributional pattern, and specifies the microdata design needed to test mechanisms.

That narrower contribution matters for public communication, but it is not yet a general-interest journal contribution. A publishable research-paper version would need to use SHED public-use microdata or comparable household survey microdata to estimate conditional gaps, uncertainty, decompositions, and links from buffer status to actual hardship outcomes.

9. What this paper does not prove

This paper does not prove that monetary conditions caused emergency-buffer weakness. It does not estimate household-level regressions, exploit policy variation, or link individual respondents over time. It does not observe actual bank balances. It does not identify whether respondents without emergency-payment capacity are also those facing higher medical, childcare, insurance, or credit costs. It does not separate inflation effects from wage growth, fiscal transfers, debt burdens, demographic shifts, or survey-composition changes.

The paper also does not claim that the $400 test is a complete resilience measure. A household might cover $400 but still be fragile under job loss. Another household might lack cash but have access to family support or low-cost credit. Conversely, some households may technically have access to credit but only at high cost, turning a small shock into a debt spiral.

These limitations define the next step. A stronger paper would combine SHED or similar microdata with income, debt, liquid assets, family structure, health exposure, insurance status, and local cost measures. It would estimate which households lack buffers after controlling for observable characteristics, and whether buffer weakness predicts borrowing, arrears, delayed care, or labor-market stress.

10. What an inferential version should estimate

The published-table exercise is a useful starting point, not the endpoint. A research-paper version should use SHED public-use microdata from 2013 through 2025 and estimate weighted models with survey-adjusted standard errors. A minimal specification would be:

`Yit = alphat + Xit' gamma + deltag + betag * Post2021t + epsilon_it`

where `Yit` *is either the $400 cash/equivalent payment-capacity outcome or the three-month emergency-savings outcome;* `Xit` includes income, employment, age, education, family structure, parental status, race and ethnicity, housing status, metro status, and other available controls; and `beta_g` measures heterogeneous post-2021 changes for groups of interest. The goal would not be to assign causal meaning to race, education, or parenthood. It would be to distinguish raw subgroup gaps from conditional differences and to quantify which parts of the 2019--2021 improvement and 2021--2025 reversal reflect composition versus within-group changes.

A stronger version should also link buffers to actual shock and hardship outcomes observed in SHED. The 2025 survey reports major unexpected expenses during the prior year, including vehicle repair or replacement, home or appliance repair, and unexpected medical expenses. The microdata version should ask whether adults without emergency-payment capacity are more likely to report bill-payment difficulty, borrowing, selling assets, skipped medical care, food insecurity, credit-card repayment difficulty, arrears, or other hardship after shocks. That link would move the paper from a published-indicator chartbook toward a testable household-finance contribution.

A still more ambitious version could exploit quasi-experimental variation: pandemic transfer exposure, Child Tax Credit eligibility, state unemployment-insurance generosity, local inflation or rent shocks, gas-price exposure, or interest-rate-sensitive debt exposure. Those designs would let the project ask which shocks and policies changed household liquidity, for whom, and with what consequences.

11. Conclusion

The emergency buffer is the household shock absorber, but this draft measures it only through published adult-level SHED indicators. In 2025, 37 percent of adults could not cover a $400 emergency expense with cash or its equivalent, and 45 percent lacked three months of emergency savings. Those averages conceal much larger gaps for younger adults, parents with children, adults with less education, and Black and Hispanic adults.

The post-2021 pattern is also important. Emergency-buffer measures improved into 2021 and then retreated or plateaued. By 2025, the small-shock cash measure was back to its 2019 level, while the larger emergency-savings measure remained below its 2021 high. This suggests that the temporary improvement in household liquidity did not become a durable broad-based buffer.

The practical lesson is straightforward. The disciplined claim is narrower than a causal theory of monetary stress. Public SHED indicators show that emergency-payment capacity improved through 2021, then returned to roughly its pre-2021 level, while large unconditional gaps remained across education, race and ethnicity, age, and parental status. That is enough to motivate deeper microdata work. It is not enough, by itself, to identify mechanisms. The emergency buffer is not the whole story, but it is where many household crises begin.

Data appendix and replication note

The source values are transcribed from Federal Reserve SHED public data visualization table pages for unexpected expenses and emergency savings, last updated May 13, 2026. The Federal Reserve data page also provides official public-use CSV microdata files by survey year, including `SHED`*`public`*`use`*`data`*`2025_(CSV).zip`, plus codebooks. The replication workflow rebuilds the published-table panel, processed tables, and figures from the public SHED source values.

The published-table approach makes the draft transparent, but it is not a complete replication package and is not a substitute for microdata analysis. A public release should include raw downloaded source files where licensing permits, processed CSVs, code, data dictionaries, version information, and generated tables and figures. Future work should use SHED microdata, Survey of Consumer Finances public data, FINRA National Financial Capability Study data, or related household surveys to estimate multivariate and distributional models of emergency-buffer adequacy.

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